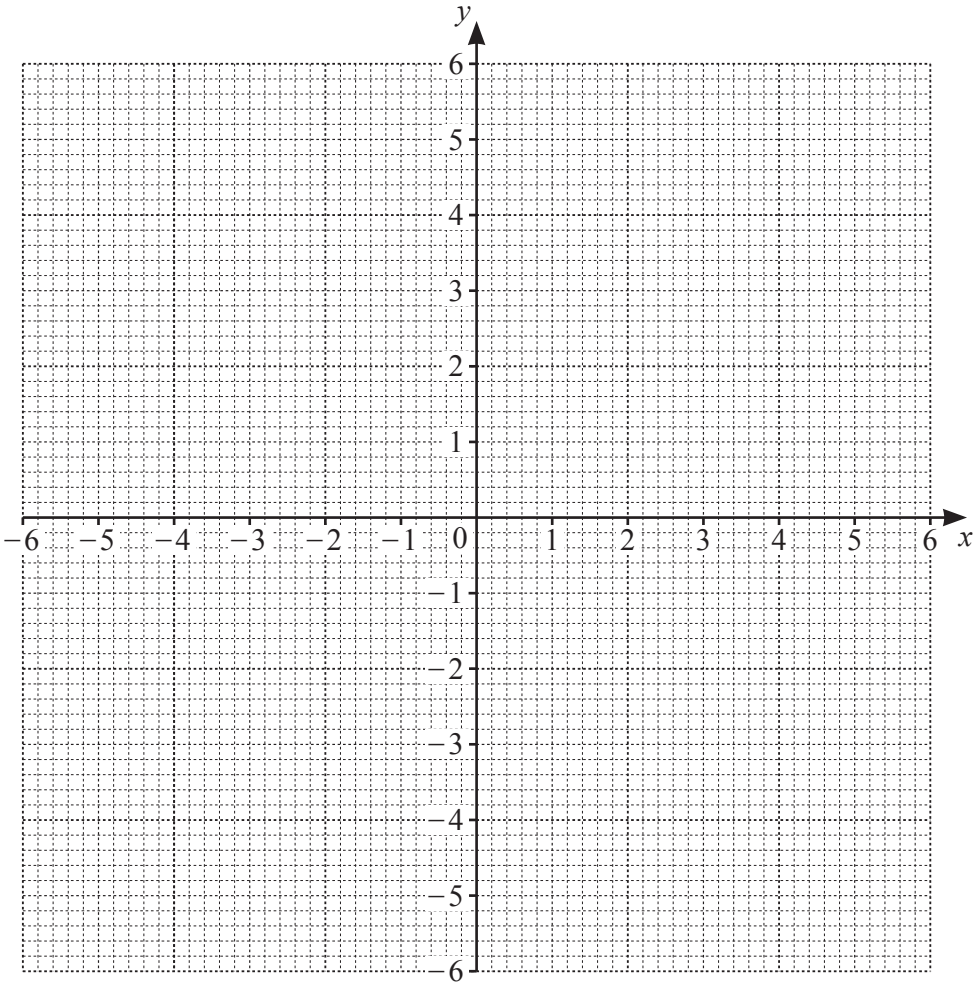
0580_s22_qp_32_Q2

2 **(a)** **(i)** Complete the table of values for $y = \frac{-6}{x}$.

x	-6	-4	-3	-2	-1.5	-1		1	1.5	2	3	5	6
y	1		2	3		6		-6		-3	-2		-1

[3]

(ii) On the grid, draw the graph of $y = \frac{-6}{x}$ for $-6 \leqslant x \leqslant -1$ and $1 \leqslant x \leqslant 6$.



[4]

(iii) Write down the order of rotational symmetry of the graph.

..... [1]

(iv) Write down the equation of each line of symmetry of the graph.

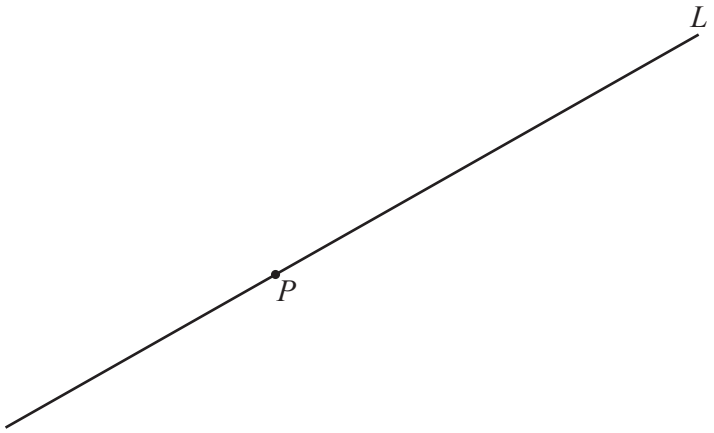
..... and ..... [2]

(v) On the grid, draw the line $y = 2.5$. [1]

(vi) Use your graph to solve the equation $\frac{-6}{x} = 2.5$.

$x =$ ..... [1]

(b)



Draw a line that passes through the point P and is perpendicular to line L. [1]

(c) Find the equation of the straight line that

- is parallel to the line $y = 3x + 5$

and

- passes through the point (1, 7).

Give your answer in the form $y = mx + c$.

$y =$ ..... [2]